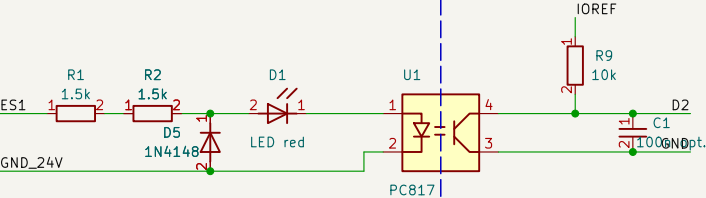


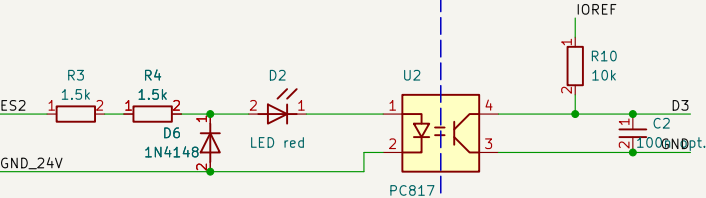
Safety inputs: 2 e-stops + 2 lidars, 24 V through PC817 – shield for Arduino UNO R4 WiFi / UNO Q

E-stop loops: $2 \times 1.5\text{ k} \rightarrow 6.9\text{ mA}$ (109 mW per resistor at 28.8 V). Lidar inputs: $2 \times 2.2\text{ k} \rightarrow 4.7\text{ mA}$ (75 mW). All resistors 1/4 W.
No current (button pressed, zone occupied, wire cut, 24 V lost) $\rightarrow$ pin HIGH = STOP. F1 protects the 24 V wires going out to the buttons.
GND_24V and GND are NOT connected: this is the isolation barrier. Pull-ups go to IOREF (5 V on UNO R4 WiFi, 3.3 V on UNO Q), never to 5V.
C1–C4 are OPTIONAL: fitted, an opening shorter than about 1.5 ms is not seen by the Arduino, and every stop is seen about 1.6 ms later.

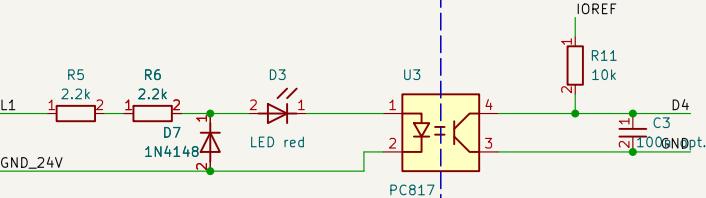
E-stop 1: button released (contact closed) $\rightarrow$ D2 LOW, LED lit



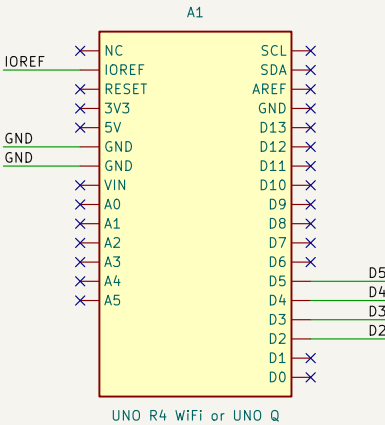
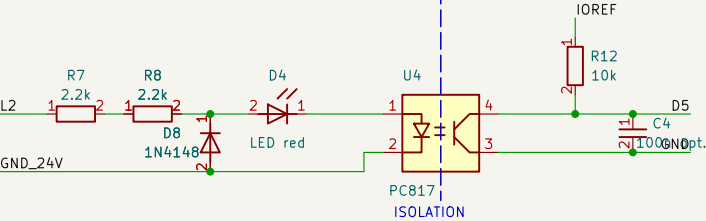
E-stop 2: button released (contact closed) $\rightarrow$ D3 LOW, LED lit



Lidar 1: output at 24 V (zone clear) $\rightarrow$ D4 LOW, LED lit



Lidar 2: output at 24 V (zone clear) $\rightarrow$ D5 LOW, LED lit



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Sheet: /
File: pc817_bench.kicad_sch

Title: Safety inputs: 2 e-stops + 2 lidars

Size: A4 Date: 2026–10–02

KiCad E.D.A. 10.0.6

Rev: 0.8

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